JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU& RTMNU

Education to Eternity

B. Tech Summer / winter2024-25
Class Test- II

Branch: CSE/AI **SEM: -III**
Subject:Numerical Methods and Discrete Mathematics
Subject Code: CS/AI3M001 **Marks: 20**
Date of Exam: 03/10/2024 **Time: 1 Hr.**

Instructions:
1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able to:(Listed Below CO & Mention it on question)
CO1: Describe the concept of Numerical methods, Bisection method and Newton-Raphson Method and their convergence, Gauss Elimination method and Crout's method.
CO2: Illustrate the concept of mathematical expectation, variance and standard Deviation, moment and moment generating function, characteristic function, conditional expectation, Varianceandmoment.
CO3: Apply the concept of joint probability and random variable, probability function Of CRV, joint probability function of DRV, joint distribution functions of DRV.
CO4: Analyse the problem by using the concept of linear dependence and independence of vector, Consistency of homogeneous and non-homogeneous equation.
CO5: Evaluate the problem based on the concept of Eigen values and Eigen vectors, Sylvester'stheorem, Quadratic form, Canonical form, differentiation & integration of matrices.
CO6: Create the new concept by using the theory of Group, group, Subgroups, Homomorphism, cosets, Lagrange's theorem and Normal subgroup.

SECTION A

Q. No.	(All Questions are compulsory).	Bloom's Level &CO	Marks
Q.1	Define characteristic equation.	L1/CO1	1
Q.2	Recall the term percentage Error.	L1/CO1	1
Q.3	Tell the formula for Variance .	L1/CO1	1
Q.4	Define: Random Variable.	L1/CO1	1

SECTION B

Q.No.	Solve answer in brief (All Questions are compulsory).		
Q.5	Interpret the formula for Mathematical ExpectationE(X) for DRV & CRV.	L2/CO2	2
Q.6	Illustrate the condition for Gauss- Seidel Method.	L2/CO2	2
Q.7	Outline the eigen values of matrix $\begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$	L2/CO2	2
Q.8	Identify the interval in which root is present for the equation $x^5 - x - 10 = 0$.	L3/CO3	2

SECTION C

Q. No.	Solve any two Questions		
Q.9	Let X be a random variable with probability distribution as follows:	L3/CO3	4

X	1	2	3
P(x)	0.2	0.5	0.3

Solve: 1) E(X) 2) $E(X^2)$ 3) $E(X^3)$ 4) $Var(X)$

Q.10	Let X be a random variable with density function	L4/CO4	4

$$f(x) = \begin{cases} \frac{x^2}{2} & , \ 0 \le x \le 2 \\ 0 & , \ otherwise \end{cases}$$

Inspect : 1) $E(X)$ 2) $E(X^2)$

Q.11	Inspect the roots of $f(x) = 4e^{-x}sinx - 1$ in the interval [0, 0.5] up to three iterations.	L4/CO4	4